

Fitting logistic regression models

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Logistic regression model

Let $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$ be iid samples from the model

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

(where the distribution of $\mathbf{x}_i$ does not depend on $\boldsymbol{\beta}$).

How do we estimate the coefficients $\boldsymbol{\beta}$?

Maximum likelihood for logistic regression

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$$L(\boldsymbol{\beta} | \mathbf{y}, \mathbf{X}) \propto \prod_{i=1}^n f(Y_i | \mathbf{x}_i, \boldsymbol{\beta}) =$$

Maximizing

$$\ell(\beta|\mathbf{y}, \mathbf{X}) \stackrel{\text{(up to a constant)}}{=} \sum_{i=1}^n \left\{ Y_i \mathbf{x}_i^T \beta - \log(1 + e^{\mathbf{x}_i^T \beta}) \right\}$$

Differentiating:

$$U(\beta) = \frac{\partial}{\partial \beta} \ell(\beta|\mathbf{y}, \mathbf{X}) =$$

Newton's method

- ▶ Want β^* such that $U(\beta^*) = \mathbf{0}$
- ▶ Begin with initial estimate $\beta^{(r)}$
- ▶ Iterative updates:

$$\beta^{(r+1)} = \beta^{(r)} - \left(\mathbf{H}(\beta^{(r)}) \right)^{-1} U(\beta^{(r)})$$

$$U(\beta) = \frac{\partial}{\partial \beta} \ell(\beta | \mathbf{y}, \mathbf{X}) = \mathbf{X}^T (\mathbf{y} - \mathbf{p})$$

$$\mathbf{H}(\beta) = \frac{\partial}{\partial \beta} U(\beta) = \frac{\partial}{\partial \beta} \mathbf{X}^T (\mathbf{y} - \mathbf{p})$$

Re-writing the update rule

Newton's method update rule:

$$\boldsymbol{\beta}^{(r+1)} = \boldsymbol{\beta}^{(r)} + (\mathbf{X}^T \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{y} - \mathbf{p}^{(r)})$$

Weighted least squares

Suppose that we had

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$$